Along the Plausible Garden-Path : Semantic versus Syntactic Reanalysis

We interpret imperfect speech daily due to speech errors, noisy environments, imperfect turn-taking or simply ambiguous language [1,2]. To assist with comprehension, we use the context and plausibility, among other things, to make sense of language. In our study, we focus on how plausibility affects reanalysis (both syntactic and semantic).

Garden-path (GP) sentences can cause lingering misinterpretations, and this even after reanalysis is presumed to have been successfully completed [3]. The source of these misinterpretations is unclear, however, as is the metric under which reanalysis is presumed to be 'successful' [4,5]. In our study, we use GP dead-ends to probe these questions further. GP dead-ends are defined as GP sentences with no grammatical resolution. In our study, the obligatorily transitive subordinate verb in an NP/Z structure eventually leads to a missing argument. In (1) (*see next page*), *the computer* is initially misinterpreted as the direct object of *fixed*. However, due to the obligatory transitivity of *fixed*, reanalysis should not be possible. However, plausibility has been shown to modulate effects of grammaticality in GP dead-ends and create a sort of grammatical illusion [4]. We compared GP dead-ends to classic GP NP/Z sentences to probe how people may be understanding classic GP sentences that, though ambiguous, are often assumed to lead to an eventual successful reanalysis.

We conducted two experiments : an acceptability judgment task and an eye-tracking while reading task. We used a 2x2 design ([im]plausibility x [un]grammaticality) to investigate the effects of plausibility on the processing of GP and GP dead-end sentences. Plausibility was manipulated locally by manipulating how well the critical NP fit as the direct object of the subordinate VP; globally, however, this critical NP was plausible in all conditions. A context sentence preceded the critical sentences to lend the task naturalness and establish context.

We conducted a forced-choice norming study to confirm our intuitions of plausibility. After picking out 40 critical trials (and adding 40 filler trials, half of which were grammatical), we conducted an acceptability judgment task to test whether our manipulations affected people's explicit judgments. After being presented with the context and critical sentences separately, participants (N=76) were asked to rate sentences on a scale from 1 (completely UNacceptable) to 5 (completely acceptable). An ordinal mixed effects model with maximal random effects found statistically significant effects of grammaticality and plausibility in the expected direction, but no interaction. This finding suggests that people were sensitive to both our plausibility and grammaticality manipulations, but plausibility did not modulate grammaticality.

Next, we conducted an eye-tracking while reading experiment using the same stimuli. Participants (N=48) were presented with each pair of sentences (context + critical) one at a time and then were asked a Y/N comprehension question. Importantly, the comprehension questions never queried the verb argument structure (the critical region) of the critical sentence : they either asked about the context sentence or about other aspects of the matrix clause. We reported two main measures at the critical disambiguating verb (i.e. *hummed*) : go-past times and regressions out and used Bayesian linear mixed effects models. In go-past durations, there was a main effect of plausibility (95% CrI [–42.06, –11.68]ms), but no main effect of grammaticality and no interaction. Similarly, when looking at regressions out, there was only a main effect of plausibility (95% CrI [–0.53, –0.21]ms). Additionally, there was an interaction between the two (95% CrI [–0.24, 0.01]ms). Plausibility being the only main effect suggests that people treated both the grammatical and ungrammatical GP sentences similarly. In line with the finding that plausibility can modulate effects of ungrammaticality [6], our findings suggest that we should be careful in how automatic we assume reanalysis to be, even if it's grammatically licensed, and how we even define success in the first place.

Our findings suggest that semantic and syntactic reanalysis are dissociable. We show that a presumed semantic interpretation of a sentence does not necessarily entail a correct syntactic reanalysis, and this despite the parser's ability to distinguish between the two (Exp. 1). Perhaps what matters more is a general feeling of adequacy with regards to the sentences; in our experiments, plausibility gives the illusion of grammaticality.

Example stimuli :

Context : Mary worked at an electronics store and her main responsibility was to repair various broken parts.

(1) While **Mary fixed *the computer*** *hummed* in the corner. (+PL, -GR)
(2) While **Mary cleaned** *the computer* hummed in the corner. (+PL, +GR)
(3) While **Mary fixed *the client*** *hummed* in the corner. (-PL, -GR)
(4) While **Mary cleaned** *the client hummed* in the corner. (-PL, +GR)

Figure 1 : Mean acceptability ratings (Exp. 1)

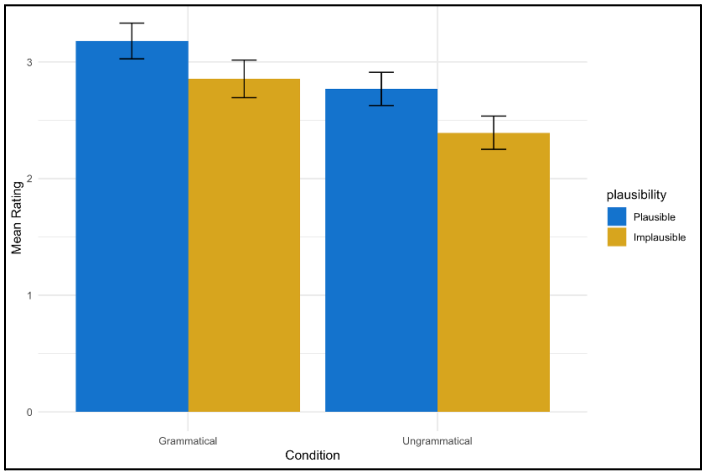
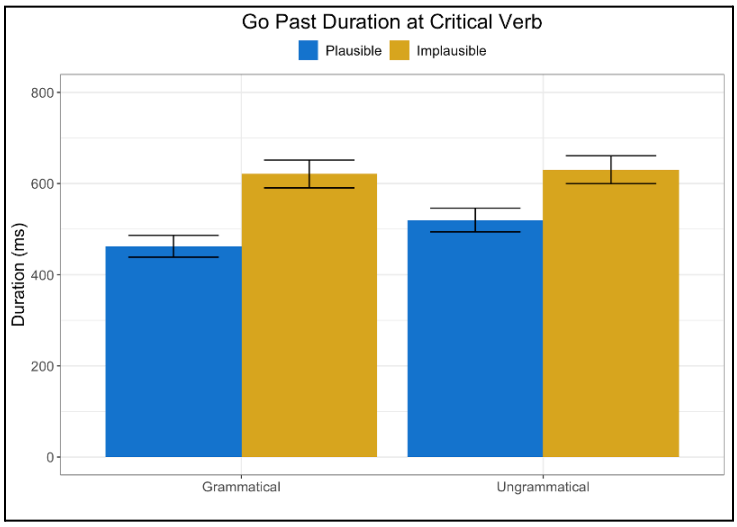
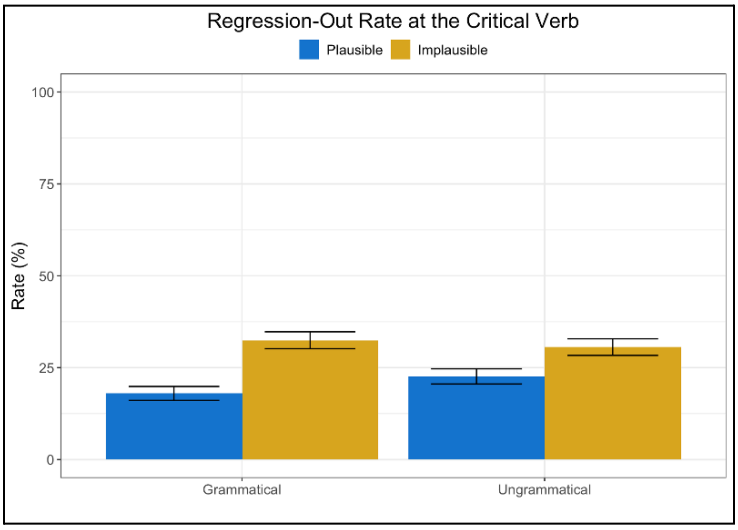


Table 1 : Ordinal mixed effects model with maximal random effects

Coefficient	*Estimate*	*SE*	*z-value*	*p*
Plausibility	0.65684	0.19724	3.330	<0.001 ***
Grammaticality	-0.72516	0.18088	-4.009	<0.001 ***
Plausibility x Grammaticality	0.06772	0.18797	0.360	0.718642





Figures 2 and 3 : Go-past durations (left) and regressions out (right) at the critical disambiguating verb

References : [1] Corps et al., “Early Preparation during Turn-Taking.”; [2] Gibson et al., “Rational Integration of Noisy Evidence and Prior Semantic Expectations in Sentence Interpretation.”; [3] Christianson et al., “Thematic roles assigned along the garden path linger.”; [4] Slatterly et al., “Lingering misinterpretations of garden path sentences arise from competing syntactic representations” ; [5] Huang & Ferreira, “What causes lingering misinterpretations of garden-path sentences: Incorrect syntactic representations or fallible memory processes?” ;[6] Hlachova & Ferreira, “Garden-path dead-ends contextualized” (abstract, HSP25)